What is intelligence?

Intelligence:

"The capacity to learn and solve problems"

particularly,

The ability to:
- Solve novel problems
- Think & act like humans
- Think & act rationally

Artificial Intelligence:

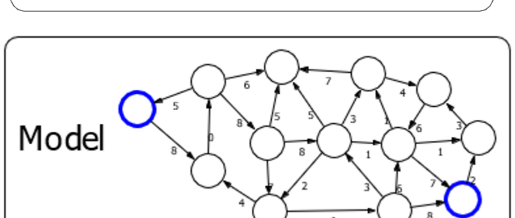
- Build intelligent entities or agents
- There are two main approaches:

1. Engineering
   - Involves a combination of mathematics and engineering, and connects to statistics, control theory, and economics.
2. Cognitive Modelling
   - The pursuit of human-like intelligence must be in part an empirical science related to psychology, involving observations and hypotheses about actual human behavior and thought processes.

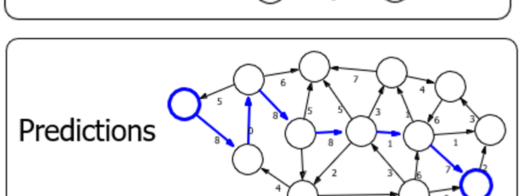
What's involved in intelligence?

1. Ability to interact with the real world
   - To precieve, understand, & act
2. Reasoning & Planning
   - Modelling the external world, given input
   - Solving new problems, planning, and making decisions
   - Ability to deal with unexpected problems, uncertai-ties.
3. Learning & Adaptation
   - We are continuously learning & adapting
   - Our internal models are being "updated".

Real world represents the complex, messy, and often overwhelming information we experience.



Model is how an AI system simplifies the real world into a structured, manageable representation.



Predictions is the final step, where the AI uses the model to find a solution or make a decision.

Artificial Intelligence as a Science

"What is the nature of intelligence & what constitutes intelligent behaviour?"

Artificial Intelligence as Engineering

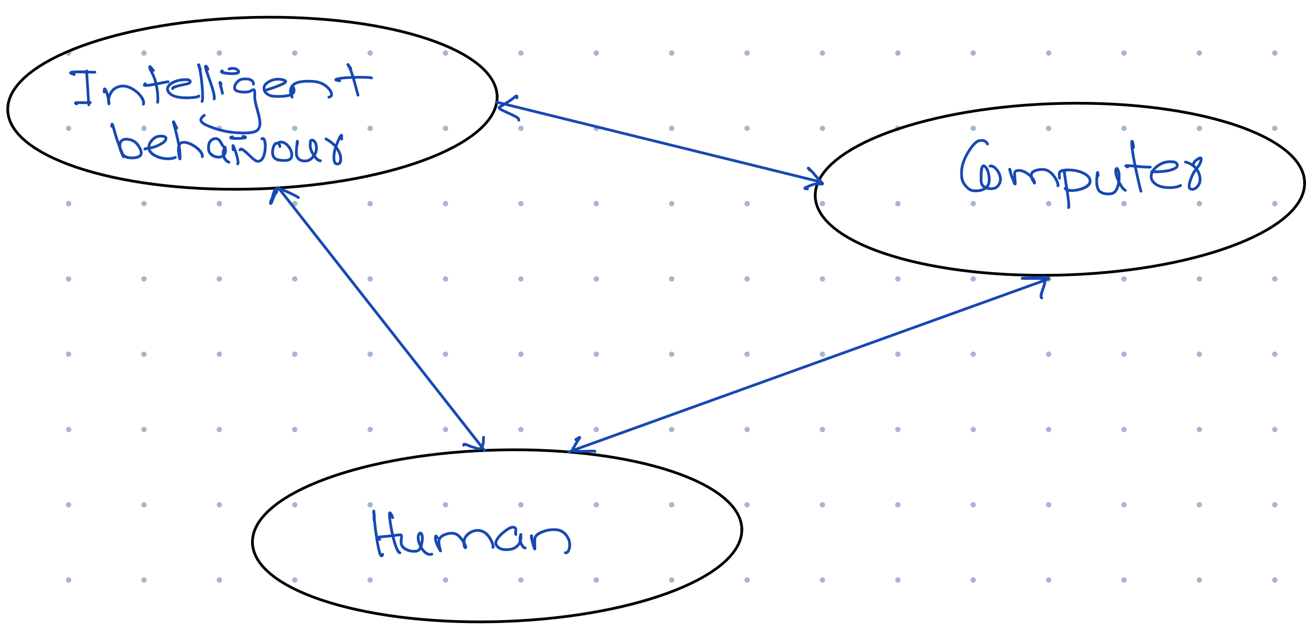
- How can we make software & machines more powerful, adaptive, and easier to use?

Examples,

- Speech Recognition
- Natural Language understanding

A simple definition:

AI is the attempt for the reproduction of human reasoning and intelligent behaviour by computational methods



What is AI?

- Four ways to define it
- A discipline that systematizes & automates reasoning processes to create machines that:

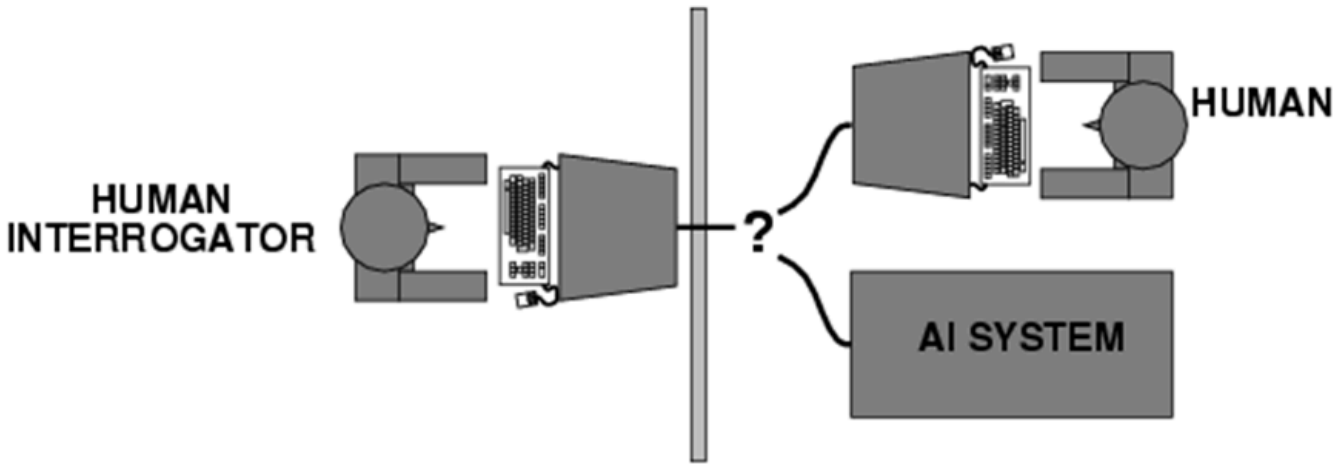
Act like humans	Act rationally
Think like humans	Think rationally

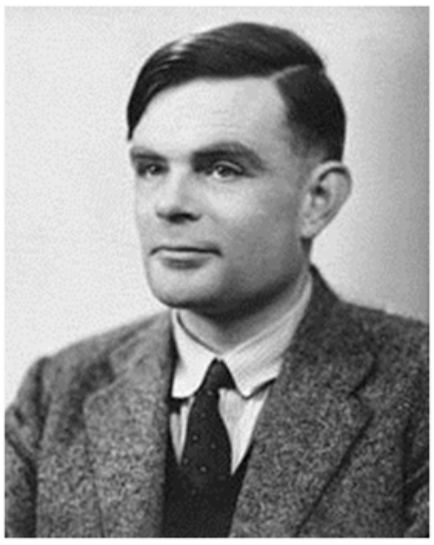
Act like humans

- The goal of AI is to create computer systems that perform tasks regarded as requiring intelligence when done by humans.
- In this we take tasks where humans are better & build a computer systems that does it automatically
  - Diagnose a disease
  - Prove a theorem ect.

↳ But humans make mistakes... do we want to replicate human imperfections.

↳ The turing test:





↳ What capabilities would a computer need to have to pass this test?

↳ NLP, knowledge representation, Automated reasoning, machine learning

↳ 2008 Competition: Each judge was given 12 minutes to conduct simultaneous, split-screen conversations with two hidden entities (human & chatbot). Elbot of Artificial solutions managed to fool 3 judges.

Examples,

↳ Eliza

↳ Simulated being a pychotherapist

↳ Successfully passed the turing test

↳ Worked by:

↳ Pattern matching

↳ Transformating input (my → your)

↳ Generating responses

↳ Using pre-defined associations with patterns

↳ An application of Turing test:

↳ CAPTCHA: Completely Automated Public Turing tests to tell Computers and Humans Apart

↳ What's wrong with turing test?

1. Variability in protocols & judges

↳ Different questioning styles affect chatbot sucess

2. Success depends on deception
↳ Passing often relies on evasive or misleading responses.

3. Cheap Tricks:
↳ Pattern matching can mimic understanding without real reasoning

↳ A better turing test?

↳ MCQs that can only be asked by humans

The trophy would not fit in the brown suitcase because it was too small.

What so small?
↳ The trophy
↳ The brown suitcase

↳ Advantage of this?

- Test can be administered and graded by machine
- Does not depend on human subjectivity
- Does not require ability to generate English sentences
- Questions cannot be evaded using verbal dodges
- Questions can be made "Google-proof" (at least for now…)

Think like humans

↳ How the computer performs tasks does matter & are inspired by human thinking.

↳ The reasoning steps are important

↳ Ability to create & manipulate symbolic knowledge

Example,
General Problem Solver (GPS) solved problems such as Tower of Hanoi

↳ Cognitive Science:
↳ Try to get "inside" our minds
Example,
Conduct experiments with humans to try & reverse-engineering how we reason, learn, remember & predict.

↳ Problems:
↳ Humans don't behave rationally
↳ The reverse engineering is hard

Act & Think Rationally:
↳ In this type of approach, the goal is to build agents that always make the "best" decision given what is available
↳ "Best" means maximizing expected value

↳ What is the impact of self-conciousness, emotions, desires on love for music, fear of dying, etc ... on human intelligence? Are they part of human intelligence

Thinking Rationally:

↳ Idealized or "right" way of thinking

↳ Logic: Patterns of arguments that yield correct conclusion when suplied with correct premises.
Example,
Socrates is a man; All men are mortal; therefore socrates is mortal.

↳ Logistict approach to AI: Describe problem in formal logical notation & apply general deduction procedures / rules to solve it.

↳ Problems with Logistic approach:
↳ Computationally expensive

- **Computational complexity:**
  - The logicist approach can lead to complex calculations that are hard to solve efficiently. Imagine trying to solve a giant puzzle with millions of pieces – it would take a long time, even with a computer!
  - Example: A self-driving car needs to make decisions quickly, but calculating every possible scenario

using logic might take too long.

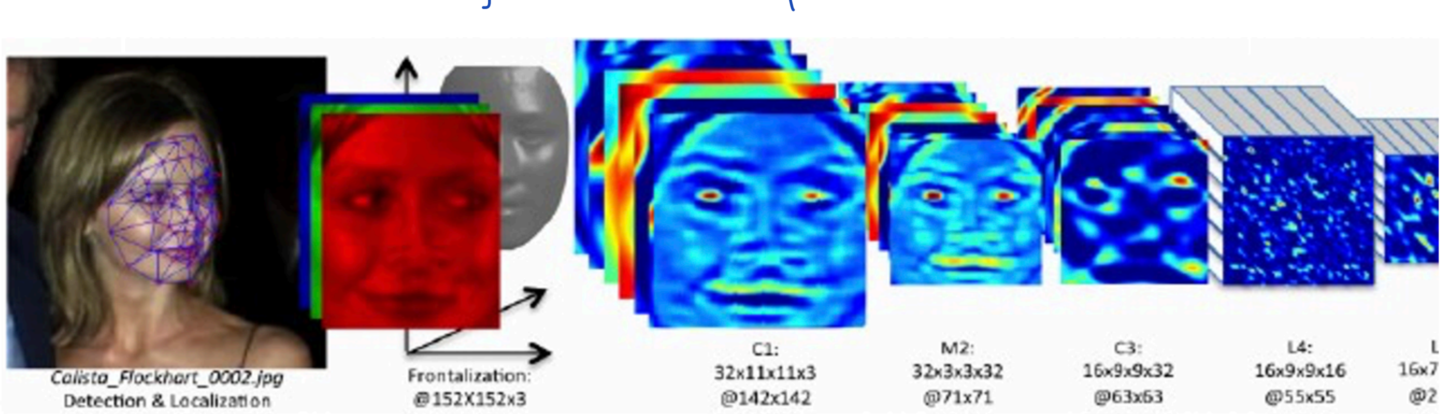
- **Describing real-world problems in logical notation:**
  - The real world is messy, and not everything can be described using strict logical rules. Some facts might be true most of the time, but not always.
  - Example: "It's sunny in Pakistan during the summer" is generally true, but not every day. Logical notation might struggle to capture these nuances.
- **Dealing with uncertainty/expected value:**
  - The logicist approach often assumes 100% certainty, but real-world decisions involve uncertainty and probabilities. We need to consider expected values and likelihoods.
  - Example: A doctor might need to decide whether to operate, given a 70% chance of success. Logic alone might not be enough to make this decision.
- **"Rational" behavior beyond logic:**
  - Human behavior is not always driven by logical reasoning. Emotions, intuition, and context play a significant role in our decisions.
  - Example:
    - Imagine a person chooses to take a longer route to work because it's scenic and reduces their stress levels, even though it's not the most efficient route. This decision might be considered rational because it maximizes their overall well-being (by reducing stress) even if it's not the shortest path.
    - In this case, the person's rational behavior (choosing the scenic route) isn't necessarily driven by logical reasoning about the most efficient route, but rather by a consideration of their emotional and psychological well-being.
    - This example illustrates how rational behavior can involve making decisions that optimize outcomes (in this case, well-being) without being based on strict logical rules or principles.

Acting Rationally:

↳ This definition of rationally only concerns the decisio-ns or actions that are made, not the cognitive process behind them.

Reflex-based models:

↳ Mapping sensory inputs directly to responses
↳ Not maintain a memory of past experiences
↳ Reacting to current stimuli without complex reasoning
Example,
↳ Linear classifiers, deep nueral networks



State-based models

- Maintain an internal state
- Use internal state information to make future decisions

Example,

- Game playing
- Robotics

- Types:

1. Search problems:
   - You control everything
   - Finding a path
   - Evaluating possible states
   - Evaluating actions

2. Markov Decision Processes
   - Decision-making in situations where outcomes are partly random & partly under control of a decision maker.

3. Adversial games:
   - Against an opponent

Sudoku

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

→

5	3	4	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6	1	5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

Goal: put digits in blank squares so each row, column, and 3x3 sub-block has digits 1–9

Key: order of filling squares doesn't matter in the evaluation criteria!

State based Possible ??

Variable-based models:

↳ Represent systems using variables
↳ Define relationship b/w variables
↳ Analyze & solve

Example,

↳ Constraint satisfaction problems
  ↳ We have hard constraints only
  Example,
  In Scheduling, one person can't be at two places at once.

↳ Bayesian Networks:
  ↳ Where variables are random & dependent on each other.